Argument Principle.

Let f be a non-constant function, analytic at z_0 with $f(z_0)=0$.
Then, for some $r>0$, f have only zero as z_0 in $B_r(z_0)$. Thus,

$$f(z)=(z-z_0)^k \cdot F(z) \text{ for some } k\in\mathbb{N},$$

and analytic function F at z_0 with $F(z_0)\neq 0$. Now,

$$f'(z)=k(z-z_0)^{k-1}F(z)+(z-z_0)^k\cdot F'(z),$$

this implies $\dfrac{f'(z)}{f(z)}=\dfrac{k}{z-z_0}+\dfrac{F'(z)}{F(z)}$.

Since $F(z_0)\neq 0$, f'/f has simple pole at z_0, with Residue k. So,

$$\int_C \frac{f'(z)}{f(z)}dz = 2\pi i\cdot k \text{ where } C \text{ is simple closed curve around } z_0, \text{ is contained in } B_r(z_0).$$

Now, let f has a pole of order n at z_0, then

$$f(z)=\frac{F(z)}{(z-z_0)^n}$$

where F is analytic at z_0 and $F(z_0)\neq 0$. Then,

$$f'(z)=-n\cdot\frac{F(z)}{(z-z_0)^{n+1}}+\frac{F'(z)}{(z-z_0)^n},$$

So therefore $\dfrac{f'(z)}{f(z)}=\dfrac{(z-z_0)^n}{F(z)}\cdot\left(-n\cdot\dfrac{F(z)}{(z-z_0)^{n+1}}+\dfrac{F'(z)}{(z-z_0)^n}\right)=\dfrac{-n}{z-z_0}+\dfrac{F'(z)}{F(z)}$

So,

$$\int_C \frac{f'(z)}{f(z)} = 2\pi i\cdot(-n).$$

obtain the two results, we obtain the following:

Thm. Let f be analytic inside and on a simple closed contour C except for finite poles in C, and $f(z)\neq 0$ on C. Then, counted with multiplicity, numbers of zeros N_f and poles P_f,

$$\frac{1}{2\pi i}\cdot\int_C \frac{f'(z)}{f(z)}dz = N_f - P_f.$$

「More observation.

Suppose that $f(z)$ is analytic and non-zero for all z on a simple closed curve C.

Set $\log f(z) = \ln|f(z)| + i\cdot \arg f(z)$

where a fixed branch for the logarithm is chosen. Then,

$$\int_C \frac{f'(z)}{f(z)}\,dz = \int_C d(\log f(z)) = \ln|f(z)|\Big|_C + i\cdot \arg f(z)\Big|_C$$

By the fundamental theorem of line integral, $\ln|f(z)|\,|_C = 0$, so

$$\int_C \frac{f'(z)}{f(z)}\,dz = i\cdot \arg f(z)\Big|_C$$

Let C' be the image of C under f. There are two case;

Thm. Rouché Theorem.

Let f and g be analytic inside and on a simple closed countor C, with

$$|g(z)| < |f(z)| \text{ for all } z \in C.$$

Then, $f+g$ and f have the same number of zeros inside C.

First Proof. By the hypothesis, $|g(z)| < |f(z)|$ on C. The triangle inequality gives

$$|f(z)| > 0 \quad \text{and} \quad |f(z)+g(z)| \geq |f(z)| - |g(z)| > 0 \qquad (z \in C).$$

Therefore, f and $f+g$ are analytic inside and on C, and have no zeros on C

By the analycity, the number of poles, $P_f = P_{f+g} = 0$.

Since f and $f+g$ cannot be zero on the countor C, write

$$f(z) + g(z) = f(z) \cdot \left(1 + \frac{g(z)}{f(z)}\right) = f(z) \cdot \phi(z) \quad \text{on } z \in C.$$

Then,
$$(f+g)'(z) = f'(z) \cdot \left(1 + \frac{g(z)}{f(z)}\right) + f(z) \cdot \left(1 + \frac{g(z)}{f(z)}\right)'$$
$$= \frac{f'(z)}{f(z)} \cdot (f(z) + g(z)) + \left(1 + \frac{g(z)}{f(z)}\right)' \cdot (f(z) + g(z)) \Big/ \left(1 + \frac{g(z)}{f(z)}\right)$$

So that
$$\frac{(f+g)'(z)}{(f+g)(z)} = \frac{f'(z)}{f(z)} + \frac{\left(1 + \frac{g(z)}{f(z)}\right)'}{\left(1 + \frac{g(z)}{f(z)}\right)} = \frac{f'(z)}{f(z)} + \frac{\phi'(z)}{\phi(z)}.$$

Let N_{f+g} and N_f denote the number of zeros of $f+g$ and f, respectively, on the domain, bounded by C. By the argument principle,

$$N_{f+g} - N_f = \frac{1}{2\pi i} \cdot \int_C \left[\frac{(f+g)'(z)}{(f+g)(z)} - \frac{f'(z)}{f(z)}\right] dz = \frac{1}{2\pi i} \cdot \int_C \frac{\phi'(z)}{\phi(z)} dz. \quad \cdots (*)$$

Since $|g(z)| < |f(z)|$, $|g|/|f| < 1$ on C, so

$$|\phi(z) - 1| = |g(z)|/|f(z)| \leq 1.$$

Taking a branch of the logarithm containing $D = \{w \in \mathbb{C} \mid |w-1| < 1\}$, then

$$\frac{\phi'(z)}{\phi(z)} = \frac{d}{dz} \log \phi(z)$$

Therefore, the integral in $(*)$ is zero, by the fundamental theorem of line integral, consequently $N_{f+g} = N_f$. ☒

Example.

Let $P(z) = z^{11} + 3z^{10} + 6z^3 + 1$. Determine the number of zeros of f in $\{z \in \mathbb{C} \mid 1 < |z| < 2\}$.

Proof). Let $f(z) = 3z^{10}$ and $g(z) = z^{11} + 6z^3 + 1$. By the triangle inequality, on $|z| = 2$,

$$|g(z)| = |z^{11} + 6z^3 + 1| \le |z|^{11} + 6 \cdot |z|^3 + 1 \le 2^{11} + 6 \cdot 2^3 + 1 \le 3 \cdot 2^{10} = 2^{11} + 2^{10} = |f(z)|.$$

Therefore, by the Rouche's Theorem, $f+g$ has same number of zeros in $|z| < 2$ to f, 10 zeros. And now, let $f(z) = 6z^3$ and $g(z) = z^{11} + 3z^{10} + 1$. Observe that

$$|f(z)| = 6 \cdot |z|^3 \quad \text{and} \quad |g(z)| = |z^{11} + 3z^{10} + 1| \le |z|^{11} + 3|z|^{10} + 1$$

We can choose $\delta > 0$ such that $0 < \varepsilon < \delta$ implies

$$|1+\varepsilon|^{11} + 3 \cdot |1+\varepsilon|^{10} + 1 < 6.$$

Therefore, for $|z| = 1 + \varepsilon$ where $0 < \varepsilon < \delta$,

$$|f(z)| = 6 \cdot |1+\varepsilon|^3 \ge |g(z)|,$$

So $P(z)$ has three zeros inside $|z| = 1 + \varepsilon$, for any $0 < \varepsilon < \delta$.

Therefore, letting $\varepsilon \to 0$, P has three zeros on $\{z \in \mathbb{C} \mid |z| \le 1\}$. Consequently it has 7 zeros in the annulus $1 < |z| < 2$. ▨

「Evaluate

$$I = \int_{|z|=2} \frac{z+2}{z(z+1)}\,dz = \int_{|z|=2} \frac{\frac{1}{z^2} + \frac{2}{z^3}}{\frac{1}{z} + \frac{1}{z^2}}\,dz = -\int_{|z|=2} \frac{f'(z)}{f(z)}\,dz$$

where $f(z) = \frac{1}{z} + \frac{1}{z^2} = \frac{z+1}{z^2}$. It has order of 2 pole at $z=0$, and has order of 1 zero at $z=-1$.

Therefore, $I = -2\pi i \cdot (1-2) = 2\pi i$.